

Weeks 2-5 Technical Progress Report -- Consolidated Catch-Up

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Note on format: weekly reporting lapsed after Week 1 while delivery continued without a pause -- four calendar weeks of shipped, tested, live-verified work had gone unreported. Rather than backfill four thin, repetitive documents, this consolidates them into one report, organised by the same weekly windows, so the actual pace and sequencing is still visible. Every claim below is checked against the current codebase, test suite, commit history and live Vercel/production configuration as of today -- not recalled from memory. Weekly cadence resumes from here.

Note on week alignment (added 16 September): a direct comparison against the original directive (COMMERCIAL PLATFORM DEVELOPMENT MILESTONES & TECHNICAL DIRECTIVE, 12 Aug-31 Oct) found this report's own "Week 2-5" labels had drifted one calendar week behind the directive's fixed windows -- e.g. this report's "Week 2" covers 24-25 Aug, which is the directive's **Week 3** (24-30 Aug) window, not its Week 2 (17-23 Aug). The section below is the first one correctly aligned: it's genuinely the directive's **Week 6** (14-20 Sep), not a relabelled "Week 5 continued."

1. What currently works -- updated status

Every row below has moved since Week 1's table; several from Partial/Not started to Working.

Journey step	Status	Change since Week 1
Customer Registration	Working	Unchanged
Organisation Onboarding	Working	Now followed by a real post-signup welcome/invite-team step (/onboarding)
Energy Asset Registration	Working	Now with technical info (manufacturer/model/serial/install date), status edit, delete, and a link to the marketplace product it was bought from
Energy Data / Monitoring	Working	New: real per-asset time-series ingestion (single + CSV batch), not just live public data + synthetic seed. Real assets now count in fault detection/rollup
Analysis & Intelligence	Working	Custom date-range analytics blended with real ingested readings, not preset-only
ANI(TM) Recommendations	Working	Closed the Week 1 gap: auto-optimize preference now has real effect (records + emails, doesn't yet dispatch hardware commands -- no hardware control exists at this stage); chat is now persistent (survives refresh) and streams token-by-token
Identification of Energy Need	Partial	Unchanged -- pilot-onboarding ROI estimates still come from a stated profile, not live telemetry
Clean-Energy Marketplace	Working	Closed the Week 1 gap: real supplier self-service accounts, product catalogue with AI-grounded Q&A, public storefront, per-product technical knowledgebase, admin verification/moderation
Relevant Product / Supplier	Partial	Unchanged -- category-matching in the pilot flow still isn't wired to ANI's live recommendation output

Journey step	Status	Change since Week 1
Enquiry / Quote / Commercial Action	Working	Closed the Week 1 gap: real state machine (new -> responded -> accepted/declined), buyer- and supplier-side views, email notification on new enquiry

Supporting infrastructure, updated:

- **Database:** DATABASE_URL/DIRECT_URL are now live in Vercel Production (confirmed via `vercel env ls`, set ~6 and ~24 days ago respectively) -- **Week 1's single largest blocker is resolved.** Two hand-authored migrations (chat_messages, the VendorLead state-machine columns) were written against no live DB at the time and need a verified `prisma migrate deploy` run confirmed against production -- flagged in Section 4, not assumed done.
- **LLM provider:** switched from Anthropic to OpenAI (OPENAI_API_KEY also live in Production) behind one shared client (`src/lib/server/llm.ts`), replacing three near-duplicate call sites.
- **Design system:** the entire app (all 24 pages) is now rebuilt on shadcn/ui primitives with the AI-generated-looking visual tells removed (pulsing-glow Sparkles assistant icon, purple/blue gradient tokens, emoji in UI copy), and -- as of today -- dark mode is the app's default theme, not a two-block accent on an otherwise white app.
- **Testing foundation:** 359 tests (up from 154), coverage thresholds unchanged and still enforced (99% statements / 85% branches / 100% functions).
- **Global market expansion:** `Org.market` (uk/eu/us) now exists end-to-end -- a real EIA-backed US region connector (derived carbon intensity from fuel-mix + monthly retail pricing, honestly `null` where the US has no equivalent to a UK metered/sub-hourly figure), and every market-dependent function in `providers.ts` (dashboard metrics, map pins, forecast, comparison, smart-charging window, price curve, optimisation) now dispatches to the calling org's actual market via `getRegionProviders()` instead of always UK. EU still falls back to the UK data set pending the ENTSO-E token. Not yet live-populated: EIA_API_KEY still isn't set in Vercel Production, so a `market: "us"` org gets honest `null`s (graceful synthetic fallback), not real numbers, until that key is added -- a five-minute, no-wait task, see Section 4.
- **Language globalization (foundation):** locale-prefixed routing (en/de/fr/es) via `next-intl`, shared chrome (Sidebar/TopBar) and the login page fully translated, auth-redirect round trip (including its `from` param) verified locale-safe in all three non-English languages. Remaining ~24 pages, ANI's dynamic narration text, and native-speaker review of the DE/FR/ES translations are explicit follow-on, not done this pass.
- **Two independent adversarial code-review passes** were run this window over previously-unreviewed commit ranges and turned up real, shipped fixes -- not just style nits (Section 2).

2. What has been improved, by week

Week 2 (24-25 August) -- Customer/Organisation, Asset, Marketplace, Supplier foundations

The single densest window of the four: eight feature commits closing most of the remaining Week 1 gaps in one push.

- Settings' Organisation/Profile/Preferences panels went from partially fake (a "Save" button that only showed a toast) to fully real and persisted.
- Team management completed: role change and removal, guarded against zero-admin orgs and self-action.
- Asset registration gained technical fields, status editing, deletion, and a link to its purchased-from marketplace product.
- **Real time-series energy ingestion shipped** -- the actual data pipeline real customer telemetry will flow through (single reading or CSV batch), replacing the previously synthetic-only Monitoring data. A real org-isolation bug (any org could read any other org's readings for a known asset id) was caught in testing and fixed before shipping.
- Monitoring's fault detection and rollup now include real user assets, not just the synthetic engine fleet.
- **Supplier self-service portal shipped** (Week 8 scope, pulled forward) -- a genuinely separate account model and JWT kind from the customer/admin/operator/pilot model, not a bolted-on role, specifically so a supplier session and a

customer session can never be confused.

- Per-product technical knowledgebase shipped, feeding ANI's product Q&A so it can resolve support questions directly and say so plainly when it can't, rather than guessing at a physical fix.
- Bulk CSV readings import, real analytics blending, marketplace admin (verify supplier / delete listing), and enquiry-email notifications closed out the window.

Week 3 (31 August - 1 September) -- Deeper analytics, scheduled reports, persistent chat, procurement workflow

- Auto-optimize preference now takes real effect (records the decision, sends a one-time confirmation email) -- closing a Week 1 gap where the toggle existed but nothing read it.
- Analytics gained custom date ranges blended with real per-org data, not presets-and-synthetic-only.
- **Scheduled, automatically-emailed reports shipped** -- a daily cron endpoint (guarded by CRON_SECRET, currently unset -- see Section 4), PDF attachment via the existing Resend integration.
- **Persistent, streaming ANI chat shipped** -- conversations now survive a page refresh (new chat_messages table) and narration streams token-by-token instead of arriving as one blob.
- An 8-bug code-review pass over the week's own commits caught and fixed real issues before they were reported here, not after: a settings toggle silently reverting to stale defaults, a double-counted reading at a comparison-window boundary, an analytics end-date that silently excluded the whole selected day, a chat-persistence failure that was discarding an already-delivered answer, two race conditions that could send duplicate notification emails, and a timing side-channel on the cron endpoint's secret comparison.
- **Marketplace procurement workflow shipped** -- "Contact Vendor" is no longer a dead-end lead-capture form; it's now a real state machine (new -> responded -> accepted/declined) with buyer- and supplier-side views.

Week 4 (7-10 September) -- Security hardening, LLM migration, full design-system rebuild

- Two further adversarial-review passes over previously-unreviewed commit ranges found and fixed: a lead-response race that could reopen an already-decided procurement enquiry, a real read-only-role (pilot) enforcement gap across six mutation routes, a platform-vs-org admin authorization gap that let any self-signed-up customer moderate other tenants' marketplace listings, a stored-HTML-injection vector across four email types, and an asset-date-clearing crash.
- Three more real bugs (invite flow silently stranding accounts when email isn't configured, two auth-path race conditions on signup and preference updates) were found and fixed in a fourth review pass.
- **LLM provider switched from Anthropic to OpenAI**, consolidating three near-duplicate raw-fetch call sites into one shared, tested client.
- **Full design-system migration**, in four verified, deployed batches: shadcn/ui foundation and shared chrome, shared component library (Panel/StatusPill/PageHeader/KpiCard, cascading to ~15-17 pages), then all 24 individual pages rebuilt in four batches (highest-visibility pages, core dashboard tabs, marketplace, auth/settings/supplier portal) -- each batch independently built, tested, and browser-verified before the next started. This also removed every remaining "looks AI-generated" visual tell (the Sparkles-icon glow, purple/blue gradient tokens, emoji in copy) per explicit direction, replacing them with the app's own navy/emerald brand identity.

Week 5 (14-15 September) -- Logo rebrand, dark-mode-default reskin, global expansion, i18n foundation

- Replaced the previous gradient/sparkle logo mark with a new monochrome wordmark across every usage (favicon, sidebar, marketing/auth pages), including generating dark- and light-background variants from the single supplied artwork.
- **Made dark mode the app's default theme.** The .dark token set already existed in globals.css from the shadcn migration but was never activated -- this turns it on globally, fills the gaps that were blocking it (dark-aware KPI tint colours, chart colours, dark: variants on every pale accent badge/callout), and fixes two real bugs found while

verifying live in-browser (a carbon-intensity gauge and a forecast confidence-band mask were both hardcoded near-white, rendering as a blown-out white shape against the new dark background). Deployed and verified.

- **Fixed ANI over-sharing unrequested data.** `selectCapabilities()` previously always bundled a second capability into every answer regardless of how weakly it matched the question; it now only bundles on a genuine score tie, and the narration prompt was rewritten to explicitly stay on-topic. ANI now answers what's actually asked instead of volunteering extra data.
- **US region real-data integration shipped** (EIA-backed carbon intensity + retail pricing, `src/lib/data/regions/us.ts`), then `providers.ts` **wired to dispatch by the calling org's market** rather than hardcoding UK everywhere -- every market-dependent dashboard/forecast/optimisation function now resolves the org's real market via `resolveTenant()` and routes through the region registry, with per-region `data_sources` labels (a US org's badges now correctly say `eia_retail_price/eia_fuel_mix`, never a UK product name for non-UK data). 8 new tests directly exercise the dispatch (a US-market org gets US-mocked data, an org with no id still defaults to UK unchanged). Known gap, not fixed this pass: ANI's own direct carbon/pricing/generation-mix chat capabilities and two unused `/api/grid/carbon/`/`/api/grid/pricing` routes still bypass this dispatch and import the UK module directly -- flagged as follow-on rather than silently left inconsistent.
- **Language globalization foundation shipped** -- next-intl locale-prefixed routing (`en/de/fr/es`) across every user-facing route (`/api/*` unaffected), shared chrome (Sidebar/TopBar) and the login page fully translated, the auth-redirect flow (including its `from` param) fixed to reason in locale-relative terms and verified safe in all three non-English languages via a full sign-out/redirect/sign-in round trip. Deployed and verified.

Week 6 (16 September, in progress) -- ANI(TM) Intelligence Foundation: model evaluation

The directive scopes this week to Jeff (Lead) and Amirreza (Potential Energy AI Lead), with "Data/Backend Developer" as support -- Amirreza's engagement is still subject to the technical assessment referenced in Section 4, so this window's contribution is scoped deliberately to the support role, not to architecture decisions that are genuinely Jeff's/Amirreza's to make.

Of the directive's six named Week 6 focus areas (technical framework, data-to-intelligence pipeline, model integration, analytical logic, recommendation framework, **model evaluation**, intelligence outputs), the first five substantially already existed -- the one genuine gap was **model evaluation**: nothing anywhere in the codebase measured whether ANI's forecast is actually accurate. Both `Insight.confidence` and `Forecast`'s band width were (and for insights, still are) functions of sample size, never of measured predictive accuracy.

- **Shipped:** `src/lib/ani/evaluation.ts` -- genuine walk-forward backtesting of the real production `forecast()` function (not a reimplementation of it): at each historical cutoff date it forecasts using only data available up to that point, then scores the prediction against what actually happened, reporting MAE/MAPE/confidence-band coverage per lead-time horizon. Wired through a new provider function, a new GET `/api/ani/forecast-accuracy` endpoint, a "Forecast Accuracy" panel on `/forecasting`, and a new ANI assistant capability so a user asking "how accurate is the forecast?" gets an evidence-based answer instead of a repeat of the same sample-size-derived confidence figure. Live-verified: 1-day-ahead forecasts backtest to ~3.6% mean error, landing inside the forecast's own band ~97-100% of the time -- notably **higher** than the page's stated "80% confidence band" target, meaning the band is currently conservative rather than miscalibrated the other way. 9 new tests, full coverage gate still passing (99.01%/88.85%/100%/99.78%).
- **Explicitly not attempted:** calibrating `Insight.confidence` (`capacity_risk` / `rising_demand` / `underutilised` / `recurring_fault`). Those insight types have no automatic, unambiguous "did it happen" ground-truth signal in this synthetic dataset -- scoring them would mean inventing a success criterion, which this codebase's own data sources deliberately avoid doing (see the honest-null pattern in `src/lib/data/regions/us.ts`). Designing what "correct" means for a rule-based insight is real Jeff/Amirreza architecture work, not something to fabricate as a backend task.
- **Still not started:** the actual Jeff/Amirreza-led architecture review of the existing ANI framework the directive's Week 6 wants (a deliberate design pass, not just an accuracy measurement bolted onto what already exists) -- and Amirreza's own technical assessment/engagement, which several of Week 6/7's other focus areas depend on.

3. What remains

Mapped against the directive's Six Core Development Layers, updated from Week 1:

- **Layer 1 (Core Platform):** Materially stronger. Auth, multi-tenancy, and the API surface (now ~55+ routes) are solid and adversarially reviewed. The database is live in production (Week 1's blocker resolved). Remaining: confirm the two pending migrations are actually applied against production (Section 4), and the branch-strategy/CI-governance question from Week 1 is still open.
- **Layer 2 (Energy Monitoring & Data):** Real per-customer telemetry ingestion now exists (single reading + CSV batch, feeding Monitoring/Analytics/the dashboard) -- this was Week 1's stated "Week 4 scope" and has shipped. **New this window:** the platform is no longer UK-only by construction -- a real US region connector exists and `providers.ts` now dispatches every market-dependent function by the org's actual market. Not yet live: `EIA_API_KEY` isn't set in Production yet (Section 4), so a US org degrades gracefully to synthetic rather than showing real numbers today. Not yet built at all: a live push/pull integration to an actual third-party hardware or metering API (today's telemetry path is manual entry or CSV; no webhook/polling connector exists) -- this is a separate gap from the market-data work above.
- **Layer 3 (ANI(TM) Intelligence):** Recommendations are grounded in real platform data end-to-end, including product Q&A now grounded in per-product manuals *and* a technical knowledgebase, with an explicit line in the system prompt between strictly-grounded product facts and general technical knowledge. Auto-optimize now has real (non-hardware) effect. **New this window:** ANI no longer volunteers a second, weakly-related answer alongside the one actually asked for -- it now answers precisely what's asked; and the forecast model now has genuine backtested accuracy measurement (Week 6 section above) rather than an unvalidated confidence figure. Not yet connected: ANI's recommendation output still isn't wired to the marketplace's *live* product catalogue (it recommends a category, not a specific real listing); its own direct carbon/pricing/generation-mix chat capabilities still hardcode UK regardless of the asking org's market (Section 4). Still open: the directive's own Week 6 architecture review (Jeff/Amirreza) of the ANI framework itself, and calibrating Insight.confidence -- both genuinely need Jeff's/Amirreza's design input, not a backend-side fix.
- **Layer 4 (Marketplace):** No longer preview-only. Supplier self-service, product catalogue, technical knowledgebase, and a real procurement state machine (enquiry -> response -> accept/decline) all shipped and are adversarially reviewed. Remaining: no payment/transaction capability (explicitly out of scope so far), and the "relevant product" matching gap noted under Layer 3.
- **Layer 5 (Clean-Energy Integrations):** Still the least-started layer. The ingestion *path* now exists (Layer 2), but the open question from Week 1 -- webhook vs. polling vs. manual as the primary pattern for a *real* third-party connection -- is still unresolved and still needs Jeff's input (Section 5), since Layer 5 work should build toward whatever ANI's recommendation pipeline expects, not be decided independently. (The US/EU market-data work above is a different axis -- which region's grid/pricing data an org sees -- not this layer's "connect a real customer asset" question.)
- **Layer 6 (Commercial Customer Experience):** **New this window:** a locale-prefixed i18n foundation (`en/de/fr/es`) shipped -- shared chrome and the login/auth flow work correctly in all four languages. This is genuinely CX-adjacent backend enablement, not the CX/CI team's own remit (customer discovery, validation, usability testing), which is otherwise unchanged from Week 1.

4. Technical blockers

Three of the five items originally flagged here are resolved; items 3 and 4 remain and are not resolvable without Nathan/Collins' direct action (account creation, a billing/org decision). Two new items (6, 7) surfaced this window as the global-expansion work matured -- called out explicitly below rather than left ambiguous.

1. ~~Two migrations needed a verified `prisma migrate deploy` confirmation against production.~~ -- **resolved.** Checked directly against the live database: both the `chat_messages` table (persistent ANI chat) and the `VendorLead` state-machine columns (procurement workflow) are present and correct. Persistent chat and procurement state are confirmed actually working in production, not just in dev/tests.

2. `CRON_SECRET` was not set in Vercel Production. -- **resolved**. A fresh secret is now set in Vercel Production; the scheduled-reports cron route will authenticate correctly from its next deploy onward.
3. `RESEND_API_KEY` is **still not set in Vercel Production**. Every email path built this window (procurement enquiry notifications, invite-flow credentials, proactive fault alerts, the on-demand digest, scheduled reports) is still running in its documented graceful-no-op mode in production -- built, tested, and silently doing nothing live. Unlike the two items above, **this isn't fixable without Nathan or Collins creating a Resend account and generating the key** -- no amount of code or CLI access substitutes for that step.
4. **Vercel project is still under a personal account**, not a Davwo Energy Ltd team -- unchanged from Week 1, and more pressing now: this window alone represents roughly four weeks of solo delivery velocity, and the directive's own multi-developer onboarding can't proceed on deploy access until this moves. **This needs Nathan's explicit call** (which Vercel team, and when) -- not something to change unilaterally given the billing/access implications. It also currently means production deploys sit behind Vercel's own deployment-protection SSO gate rather than a plain public URL -- worth factoring into the team-migration decision.
5. **Deploy itself is still a manual** `vercel --prod`, **though this is narrower than "no pipeline" made it sound**. A GitHub Actions CI workflow (`.github/workflows/ci.yml`) has in fact been gating every push/PR to main/staging with type-check + lint + full test suite + build since 22 August -- that safety net already exists and was simply under-credited in the last draft. What's still manual is the final `vercel --prod` step itself; wiring that to run automatically post-merge (or connecting Vercel's own git integration) is the remaining gap, and is worth doing once the Vercel-team move (item 4) lands rather than before, so deploy permissions land in the right place the first time.
6. `EIA_API_KEY` is **not set in Vercel Production**. The US region connector (carbon intensity + retail pricing) and this window's `providers.ts` market-dispatch wiring are both code-complete and tested, but with no key set a `market: "us"` org gets an honest `null` (graceful synthetic fallback), not real data. Unlike `RESEND_API_KEY`, EIA registration is instant (no approval wait) -- this is genuinely a five-minute task once Nathan or Collins does it, not something blocked on a third party.
7. **ENTSO-E API token hasn't landed yet**, so the eu market still falls back to the UK data set rather than having its own real connector. This needs the manual email request to `transparency@entsoe.eu` (~3 business days) sent, if it hasn't been already -- not something this session can trigger on your behalf.

5. Dependencies on Jeff

- **Architecture sign-off is still open** on everything flagged in Week 1 (self-service signup/dynamic tenants, Postgres/Prisma) *plus* everything shipped since -- most materially the supplier-portal account/JWT model (Week 2) and the procurement state machine (Week 3), since both introduce new authorization boundaries that should get the same review rigor as the two Week 1 items, not be treated as unilaterally settled just because they're tested and adversarially self-reviewed.
- **The Layer 5 "connecting a real asset" decision** (webhook vs. polling vs. manual/CSV) is more urgent now than in Week 1: the ingestion path it needs to plug into now exists and is live, so this is no longer a future-looking architecture question -- it's blocking the next concrete increment of Layer 5 work today.
- **Branch-strategy / review-gate decision** from Week 1 is still open and now more urgent given four weeks of direct-to-main velocity with no second reviewer on the team yet.

6. Dependencies on other developers

- Still none -- Developer A and Developer B have not been onboarded, unchanged from Week 1. Given the Vercel-account blocker (Section 4, item 4), onboarding either of them productively is still gated on that move happening first.
- Amirreza's energy-AI/optimisation work dependency is unchanged from Week 1, but the ANI data pipeline it depends on is now materially more stable (real ingestion, persistent chat, real optimise-preference effect all shipped) -- worth flagging that this dependency may now be closer to ready than Week 1 estimated.

7. Recommended next sprint

1. ~~Verify the two pending migrations against production and confirm CRON_SECRET is added to Vercel~~ -- **done.**
2. **Set EIA_API_KEY in Vercel Production** -- instant registration, no wait, and it's the only thing standing between the already-shipped US region connector and actually showing real US customers real data (Section 4, item 6).
3. RESEND_API_KEY and **moving the Vercel project to a Davwo Energy Ltd team** remain open, both needing Nathan/Collins' own action (Section 4, items 3-4).
4. **Send (or confirm sent) the ENTSO-E token request** to transparency@entsoe.eu -- the EU region connector is otherwise ready to build once it lands (Section 4, item 7).
5. **Get Jeff's decision on the Layer 5 ingestion pattern** -- the door for this is open now that Layer 2's pipeline is live; deciding it late means rework.
6. **Jeff (and Amirreza, once engaged) to run the actual Week 6 architecture review** -- the framework/pipeline/recommendation-framework review the directive names, plus a real answer to what "correct" means for Insight.confidence so it can be backtested the way the forecast model now is (Section 2's Week 6 note).
7. Then proceed into the remaining open product gaps: wiring ANI's recommendations to real live marketplace listings (Layer 3/4 boundary); fixing ANI's direct carbon/pricing/generation-mix chat capabilities to respect an org's market (the same fix providers.ts just got, applied to the two places that still bypass it); translating the i18n foundation's remaining ~24 pages and ANI's dynamic narration; and a live telemetry connector for Layer 5 once Jeff's pattern decision lands.